

Memory Management Techniques

Contiguous Allocation: First Fit & Best Fit Simulation

In contiguous memory allocation, the OS must satisfy process memory requests from a list of available free blocks (holes). This guide simulates two fundamental allocation strategies: First Fit and Best Fit.

1. System Memory Snapshot

Before allocating, we need an initial snapshot of the physical memory landscape (free holes) and a queue of incoming processes with their respective memory demands.

```
user@os-sim:~/mem$ cat memory_state.conf
# Initial Free Memory Blocks (Holes)

Block_1: 100 KB
Block_2: 500 KB
Block_3: 200 KB
Block_4: 300 KB
Block_5: 600 KB

# Incoming Process Queue (Size Required)

Process_P1: 212 KB
Process_P2: 417 KB
Process_P3: 112 KB
Process_P4: 426 KB
```

2. First Fit Allocation Algorithm

First Fit scans the memory from the beginning and allocates the *first* hole that is large enough. It is fast but can heavily fragment the front of the memory space.

```
user@os-sim:~/mem$ ./allocator --algo FIRST_FIT --file memory_state.conf
Initializing First Fit Allocator...

# Allocating P1 (212 KB)

-> Scanning: Block_1(100) Too small. Block_2(500) OK.
-> ALLOCATED: P1 placed in Block_2. Remaining hole: 288 KB.

# Allocating P2 (417 KB)

-> Scanning: B1(100), B2(288), B3(200), B4(300) Too small. Block_5(600) OK.
-> ALLOCATED: P2 placed in Block_5. Remaining hole: 183 KB.

# Allocating P3 (112 KB)

-> Scanning: Block_1(100) Too small. Block_2(288) OK.
-> ALLOCATED: P3 placed in Block_2. Remaining hole: 176 KB.

# Allocating P4 (426 KB)

-> Scanning: B1(100), B2(176), B3(200), B4(300), B5(183).
-> FAILED: No contiguous block large enough for P4.

STATUS: P4 is waiting. Algorithm finished with unallocated processes.
```

3. Analyzing External Fragmentation

Even though P4 failed to allocate in First Fit, let's check the total available memory left in the system to understand *External Fragmentation*.

```
user@os-sim:~/mem$ ./allocator --stat
```

```
Current Memory Map Status:
```

```
Block_1: 100 KB (Free)
```

```
Block_2: 176 KB (Free)
```

```
Block_3: 200 KB (Free)
```

```
Block_4: 300 KB (Free)
```

```
Block_5: 183 KB (Free)
```

```
Total Free Memory: 959 KB
```

```
Largest Contiguous Hole: 300 KB
```

```
# Conclusion: P4 requires 426 KB. We have 959 KB total free space,
```

```
# but because it is scattered (fragmented), P4 cannot be loaded.
```

4. Best Fit Allocation Algorithm

Best Fit scans the *entire* list and allocates the smallest hole that is big enough. This preserves larger holes for future large processes, though it creates very tiny, unusable left-over holes.

```
user@os-sim:~/mem$ ./allocator --algo BEST_FIT --file memory_state.conf
Initializing Best Fit Allocator... (Resetting Memory Map)

# Allocating P1 (212 KB)

-> Candidates >= 212: B2(500), B4(300), B5(600).
-> Best choice is Block_4 (300 KB).
-> ALLOCATED: P1 placed in Block_4. Remaining hole: 88 KB.

# Allocating P2 (417 KB)

-> Candidates >= 417: B2(500), B5(600).
-> Best choice is Block_2 (500 KB).
-> ALLOCATED: P2 placed in Block_2. Remaining hole: 83 KB.

# Allocating P3 (112 KB)

-> Candidates >= 112: B3(200), B5(600).
(B1, B2, B4 are too small)

-> Best choice is Block_3 (200 KB).
-> ALLOCATED: P3 placed in Block_3. Remaining hole: 88 KB.

# Allocating P4 (426 KB)

-> Candidates >= 426: B5(600).
-> Best choice is Block_5 (600 KB).
-> ALLOCATED: P4 placed in Block_5. Remaining hole: 174 KB.

STATUS: SUCCESS. All processes successfully loaded into memory.
```